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A New Method for Identifying Low-Resistivity Gas Zones, High-Resistivity Water Zones, and Water-Flooded Layers Based on SP and Array Induction Resistivity

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Abstract

The identification of oil, gas, water and water-flooded zones is one of the main tasks in well logging processing and interpretation, and its accuracy directly impacts hydrocarbon reservoir evaluation, reserve estimation, and development planning. Conventional fluid identification primarily relies on electrical property differences, where hydrocarbons (oil/gas) exhibit significantly higher resistivity than water zones.

With the advancement of exploration and development, the study of unconventional reservoirs such as low-resistivity gas reservoirs has become increasingly important. Compared to conventional high-resistivity gas zones, low-resistivity gas reservoirs exhibit lower resistivity, often overlapping with water zones, making traditional resistivity-based identification challenging. To address this issue, this paper proposes a rapid fluid identification method based on spontaneous potential (SP) and array induction resistivity (AT).

The workflow mainly includes the following 6 steps: 1) **Investigate** mud resistivity (R_{mf}) and formation water resistivity (R_w). 2) **Compute** array induction resistivity index (AT_{index}) to characterize resistivity differences between invaded zones and formation, enabling low-resistivity pay and high-resistivity water layer identification. 3) **Derive** SP index (SP_{index}) reflecting salinity differences to identify water-flooded zones. 4) **Amplify** fluid curve discrepancies and calculate fluid characterization index (F_{index}). 5) **Establish** fluid identification templates and cutoff values using 2D/3D crossplots. 6) **Validate** results with dynamic data for efficient fluid identification.

In the actual application at Field R in South America, this method enables rapid identification of low-resistivity gas zones, high-resistivity water layers, and water-flooded zones, with results showing strong consistency with production data. This confirms the reliability and applicability of the approach, significantly enhancing reservoir characterization and providing critical guidance for the next phase of gas field development.

Introduction

The low resistivity oil layer usually refers to its resistivity ratio between oil layer and adjacent pure water layer is less than 2 in the same oil-water system (Sun Jianmeng,1998; Zhao Fazhan,2005; Ye Xinming,2006).

With the deepening of exploration job, amount of IOIP of low resistivity reservoirs has been found all over the world (Shan Xiang,2014).

In recent years, the research focus has been on etiological mechanisms and identification technology. There are 5 types of the formation mechanisms of low resistivity oil layers: high irreducible water saturation, clay additional conductivity, country rock influence, salinity difference and multi-mechanism compound (Wang Ruihong,2016; Yang Jiao, 2014). Low-resistivity annulus can be formed by invasion of diluted cement slurry (Liao Dongliang,2014). The comprehensive identification and prediction technology of low resistivity reservoir provides effective technical support for increasing oil and gas production in Block F (Yu Wenwen, 2020), But it needs a lot of basic information. Array Induction Log; NMR; MDT are widely used in the evaluation of low-resistivity oil and gas layers, which can distinguish oil and gas from water layers (Zhao Junlong, 2011). MDT data is well used in Sudan (Yu Wenwen, 2021) for low resistivity oil field OWC analysis.

The study of water-flooded layers focuses on the law of resistivity change, well logging response characteristics, and identification methods. Based on the method of wavelet entropy , the time-frequency characteristics of resistivity and porosity curves are analyzed to amplify the information of water flooding and effectively the degree of water flooding in low-resistivity oil layers (Wang Renyi, Liu Yueqiang 2007).

The challenge is that low-resistivity oil and gas layers and flooded layers have a high degree of overlap in their well logging responses, making it difficult for traditional methods (such as resistivity-porosity cross plots) to accurately distinguish them. Fast identification of low-resistivity gas layers and water-flooded layers is of great importance.

Method

In this paper, a new method for identifying low-resistivity gas zones, high-resistivity water zones, and water-flooded layers based on SP and Array induction resistivity was proposed.

The workflow mainly includes the following 6 steps: investigate drilling mud filtrate resistivity and formation water resistivity, Array Induction Resistivity Index (ATindex) calculation, Spontaneous Potential Index (SPindex) calculation, Fluid Identification Index (Findex) Calculation, cutoff value determination and fluid type identification.

Investigate drilling mud filtrate resistivity and formation water resistivity

The difference between mud filtrate resistivity (R_{mf}) and formation water resistivity (R_w) is a critical parameter in well log interpretation, particularly for fluid identification, saturation calculation, and distinguishing hydrocarbon-bearing zones from water zones.

To accurately determine R_{mf} (drilling mud filtrate resistivity) and R_w (formation water resistivity) as critical inputs for fluid identification in well log interpretation, particularly in low-resistivity reservoirs where conventional methods fail.

R_{mf} measurement is from laboratory analysis of mud filtrate samples collected during drilling.

R_w determination is from laboratory analysis of produced water samples (if available) and SP (Spontaneous Potential) log interpretation in clean water zones.

Array Induction Resistivity Index (ATindex) calculation

Array Induction Resistivity (AT) is an advanced logging technology that provides multiple depths of investigation (DOIs) with high vertical resolution, making it indispensable for fluid typing, invasion profiling, and low-resistivity reservoir evaluation.

The resistivity amplitude difference, which captures the contrast between the invaded zone and virgin formation resistivity, is quantified by computing the difference between the AT10 and AT90 curves, termed the Array Induction Resistivity Index (ATindex).

$$ATindex = \frac{(AT10 - AT90)}{(AT10 + AT90)} * AT10$$

Spontaneous Potential Index (SPindex) calculation

The SP log records natural electrical potentials generated at formation boundaries, reflecting lithology/permeability contrasts (sand vs. shale interfaces) and electrochemical processes (ion diffusion/filtration) between fluids of differing salinity.

After performing baseline correction and standardization on the spontaneous potential (SP) curve, the amplitude anomaly value of the SP curve is calculated. This anomaly value reflects changes in formation fluid properties and can be obtained by computing the deviation between the SP curve and the baseline curve, which is referred to as the Spontaneous Potential Index (SPindex).

$$SPindex = \frac{(SP - SPmin)}{(SPmax - SPmin)}$$

Fluid Identification Index (Findex) Calculation

To enhance fluid identification in low-resistivity pay zones and water-flooded reservoirs, a new index (Findex) is proposed by integrating resistivity amplitude difference (ATindex) and SP log difference (SPindex). This method amplifies the contrast between hydrocarbons and water-bearing zones by leveraging both physical (resistivity) and chemical (ionic activity) properties of formations.

This index enables the identification of distinct fluid types (e.g., oil, gas, water) by integrating and weighting key parameters :

$$Findex = \frac{ATindex}{(1 - SPindex)}$$

Cutoff value determination

To establish fluid-type thresholds (gas, water, water-flooded zones), 2D methods (histograms, resistivity vs. SP cross-plots) and 3D cross-plots are used. 2D analysis identifies basic trends, while 3D plots enhance discrimination by integrating three log parameters (SPindex, ATindex, Findex). Test/production data calibrate thresholds. 3D cross-plots reduce ambiguity by visualizing complex relationships.

Fluid type identification

To precisely identify fluid types (gas, oil, water, water-flooded zones) using SPindex, ATindex, and Findex, we combine amplitude anomalies (SPindex) and resistivity differences (ATindex) with a weighted composite index (Findex) to enhance discrimination. Based on the qualitative analysis and 3 parameters cutoff value to determine the fluid type.

Field Application: Case Study in the R Gas field, South America

In the actual application at Field R in South America, this method enables rapid identification of low-resistivity gas zones, high-resistivity water layers, and water-flooded zones, with results showing strong consistency with production data. This confirms the reliability and applicability of the approach, significantly enhancing reservoir characterization and providing critical guidance for the next phase of gas field development.

Investigate drilling mud filtrate resistivity and formation water resistivity

The resistivity of mud filtrate was calculated according to the background investigation of the study area, and the resistivity could be known as $R_{mf} > 0.5$ ohm.m (Table.1), which was a freshwater mud filtrate according to the calculation of the mineralization degree of mud filtrate, the salinity of mud filtrate was < 5000 ppm. The mud salinity is low and the R_{mf} is high.

Rmf calculation function:

$$Rmf = \frac{0.0123 + \frac{3647.5}{TDS^{0.955}}}{1 - 0.0175 \cdot (T - 25)}$$

Table 1—Mud filtrate properties

Cl ⁻ (ppm)	Ca ²⁺ (ppm)	TDS (ppm)	Rmf (Ω·m,25°C)	Rmf (Ω·m,75°C)
1,700	80	2,880	1.41	0.68
2,200	80	3,700	1.08	0.52
1,800	120	3,090	1.29	0.62

The formation water resistivity was calculated according to the background investigation of the study area, and the water was saline water with a mineralization degree of 15000-25000ppm and a resistivity of 0.1-0.2ohm.m.

Based on drilling mud and formation water analysis, the reservoir features fresh mud (high Rmf) and high-salinity formation water (low Rw), with freshwater injection for production.

ATindex, SPindex and Findex calculation

ATindex, SPindex and Findex were calculated for fluid type identification.

In the San_Wm formation (Fig.1), while the absolute resistivities (AT10 and AT90) show no significant difference between the upper and lower sections, the resistivity amplitude difference ($\Delta AT = AT90 - AT10$) varies distinctly, indicating different fluid responses. The array induction resistivity index (ATindex) effectively distinguishes the upper low-resistivity hydrocarbon zones from the lower high-resistivity water layers. By computing the ATindex (a derived parameter amplifying resistivity contrasts), we successfully identified that:

- Upper section: Low-resistivity gas zone (minimal AT variation), The Rt(gas) is relatively low because it is affected by low Rw (because of the high salinity formation water). No mud invasion and $AT10 \approx AT90$.
- Lower section: High-resistivity water layer (ATindex increase), The Rt(water) is much lower than the Rt(mud). Mud invasion into the transition zone and $AT10 > AT90$.

In the adjacent well (right), additional discrimination was achieved through SP amplitude variations. Combined with the spontaneous potential index (SPindex), the fluid characterization index (Findex) amplifies fluid curve differences, enabling identification of the upper low-resistivity pay zones and lower water-flooded intervals. The SPindex (quantifying spontaneous potential deviations) was integrated with ΔAT to calculate a composite fluid identification parameter (Findex), enabling precise differentiation of:

- Low-resistivity gas zones (ATindex decreasing, SP negative anomaly), The Rt(gas) is relatively low because it is affected by low Rw (because of the high salinity formation water). No mud invasion $AT10 \approx AT90$, the formation water salinity is different with the mud, SP negative anomaly.
- High-resistivity water layers (ATindex increasing, SP negative anomaly), The Rt(water) is much lower than the Rt(mud). Mud invasion into the transition zone and $AT10 > AT90$, the formation water salinity is different with the mud, SP negative anomaly.
- Water-flooded intervals (ATindex increasing, flattened SP), The Rt(water) is much lower than the Rt(mud). Mud invasion into the transition zone and $AT10 > AT90$, the formation water salinity is similar with the mud, the SP shows no amplitude anomaly.

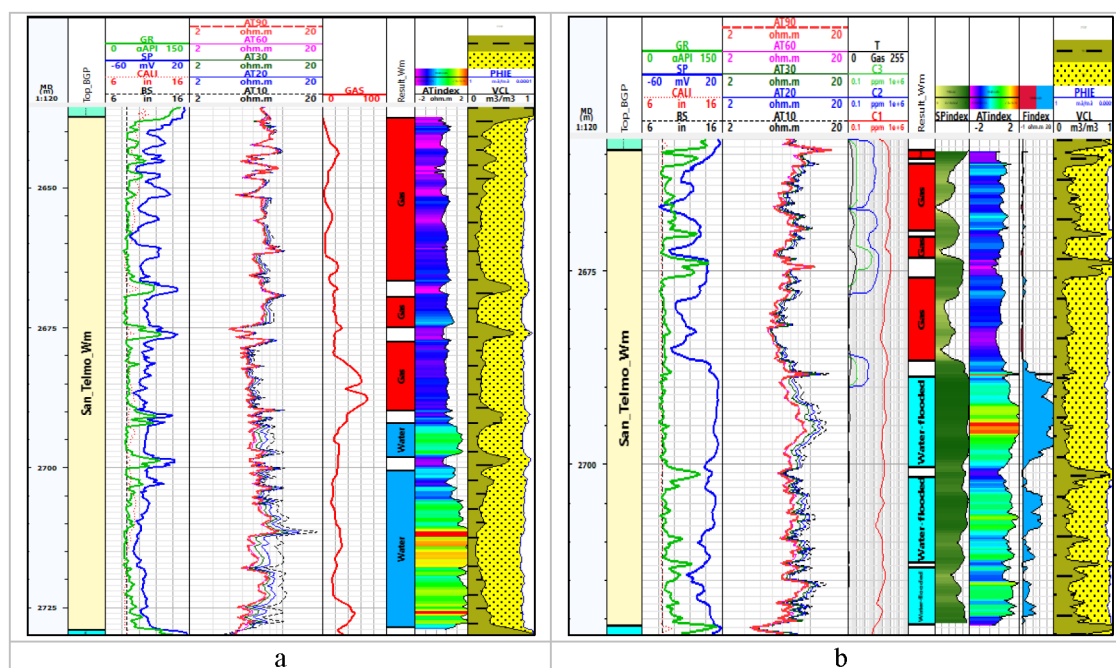


Figure 1—ATIndex, SPIndex and Findex calculation. a) The array induction resistivity index (ATIndex) effectively distinguishes the upper low-resistivity hydrocarbon zones from the lower high-resistivity water layers. b) Combined with the spontaneous potential index (SPIndex), the fluid characterization index (Findex) enabling identification of the upper low-resistivity pay zones and lower water-flooded intervals.

Cut off determination

These three parameters qualitatively differentiate low-resistivity hydrocarbons, high-resistivity water, and water-flooded zones, and based on quantitative research, the fluid identification will be more accurate.

However, the cutoff value is determined based on established production and tested intervals, and the number of samples directly influences its uncertainty. A larger number of sample points from production and tested intervals results in lower uncertainty in the cutoff value. Therefore, fluid identification should not rely solely on cutoff values but must also incorporate a comprehensive qualitative analysis of multiple logs. Moreover, cutoff values vary across different regions.

Three methodologies were employed to determine cutoff values: 2D histograms (Fig.2), 2D crossplots (Fig.3), and 3D crossplots (Fig.4).

From 2D histograms, the original curve AT90 cannot be used to identify fluid type. ATIndex can identify low-resistivity hydrocarbon zone (<1), but cannot identify water zone and water-flood zone. SPIndex can identify water-flooded zone (>0.4), but cannot identify hydrocarbon zone and water zone, Findex to determine 3 fluid types, but with overlapping.

From 2D crossplots, the original curve SP-AT90 crossplot cannot identify different fluid types. SPIndex-ATIndex crossplot can identify different fluid types, but with overlapping.

The 3D crossplot method significantly enhances fluid discrimination by amplifying parameter contrasts through ATIndex, SPIndex, and Findex, enabling more effective determination of cutoff values in three-dimensional space.

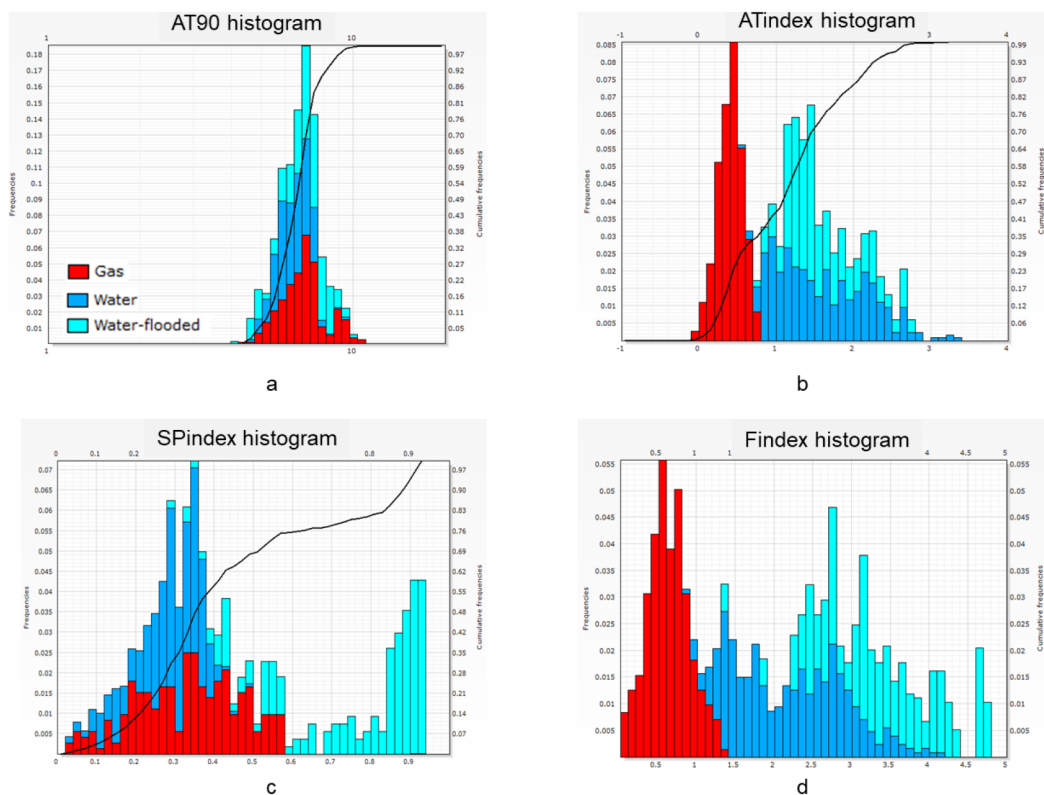


Figure 2—ATIndex, SPIndex and Findex are used to identify fluid type. a) AT90 can not be used to identify fluid type. b) ATIndex can identify low-resistivity hydrocarbon zone(<1), but can not identify water zone and water-flood zone. c) SPIndex can identify water-flooded zone(>0.4), but can not identify hydrocarbon zone and water zone, d) Findex to determine 3 fluid types, but with overlapping.

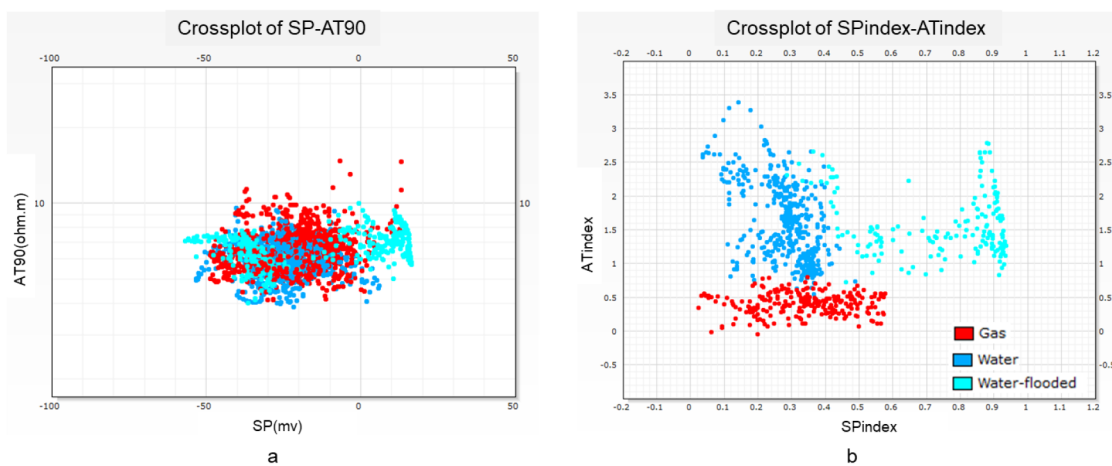


Figure 3—2D crossplots are used to identify fluid type. a) SP-AT90 crossplot can not identify different fluid types. b) SPIndex-ATIndex crossplot can identify different fluid types.

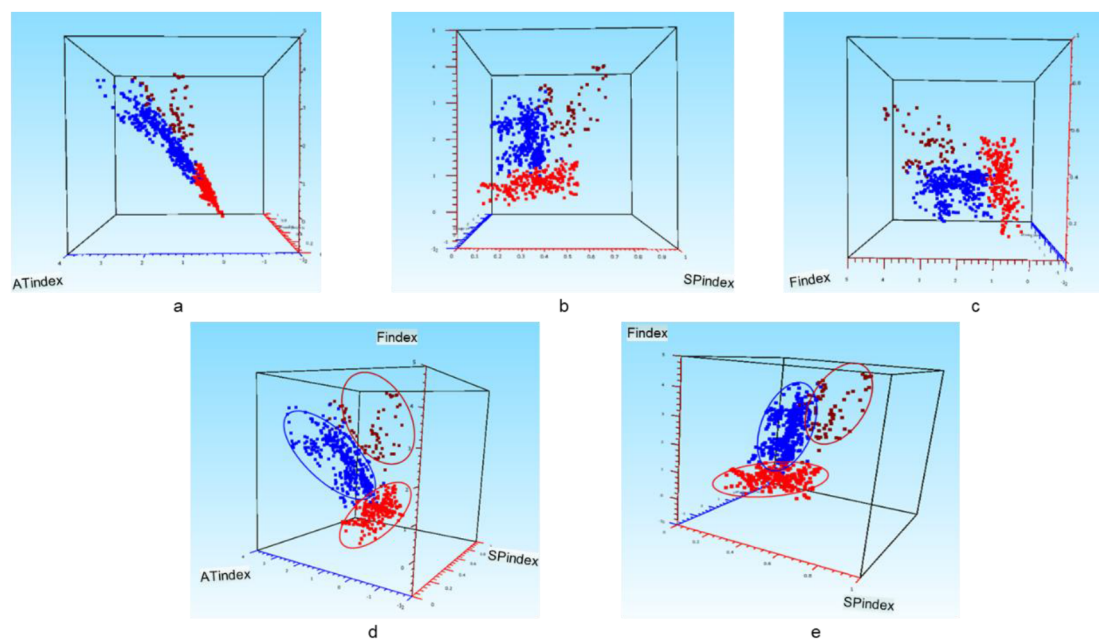


Figure 4—3D crossplots to establish fluid identification templates and determine cutoff values. Low-resistivity hydrocarbon zone (SPindex<0.55, ATindex<1, Findex<1), high-resistivity water zone (SPindex< 0.4, ATindex>1, Findex>1), and water-flooded zone (SPindex>0.4, ATindex>1, Findex>2). (red points-hydrocarbon zone, blue points-water zone, purple points-water-flooded zone)

This methodology introduces multidimensional fluid discrimination by simultaneously analyzing resistivity-based (ATindex), electrochemical (SPindex), and composite (Findex) responses through advanced 3D visualization. The enhanced cluster separation achieved in the 3D coordinate space (X-Y-Z: ATindex-SP index-Findex) provides superior fluid typing compared to conventional 2D approaches. Furthermore, dynamic cutoff optimization is enabled through interactive 3D plot rotation, allowing operators to identify optimal threshold planes that maximize differentiation between gas zones, water zones, and water-flooded formations based on production data and reservoir characteristics. This integrated approach significantly improves interpretation accuracy in complex reservoirs where traditional methods fail.

3D crossplots to establish fluid identification templates and determine cutoff values (Table.2). Low-resistivity hydrocarbon zone (SPindex<0.55, ATindex<1, Findex<1), high-resistivity water zone (SPindex< 0.4, ATindex>1, Findex>1), and water-flooded zone (SPindex>0.4, ATindex>1, Findex>2).

Table 2—Cutoff value of different fluid types

	Gas zone	Water zone	Water-flooded zone
SPindex	<0.55	<0.4	>0.4
ATindex	<1	>1	>1
Findex	<1	>1	>2

Fluid type identification

Based on qualitative and quantitative analysis, using SP, AT log curve and cutoff value range to identify fluid type. Update the well log interpretation results, validated with dynamic data (Fig.5), this approach enables rapid, efficient, and accurate fluid typing, guiding exploration and development. This workflow enables rapid, efficient, and accurate fluid identification while significantly reducing interpretation uncertainty. The standardized approach provides reliable guidance for both exploration targeting and development optimization, particularly in complex reservoirs where conventional methods often fail.

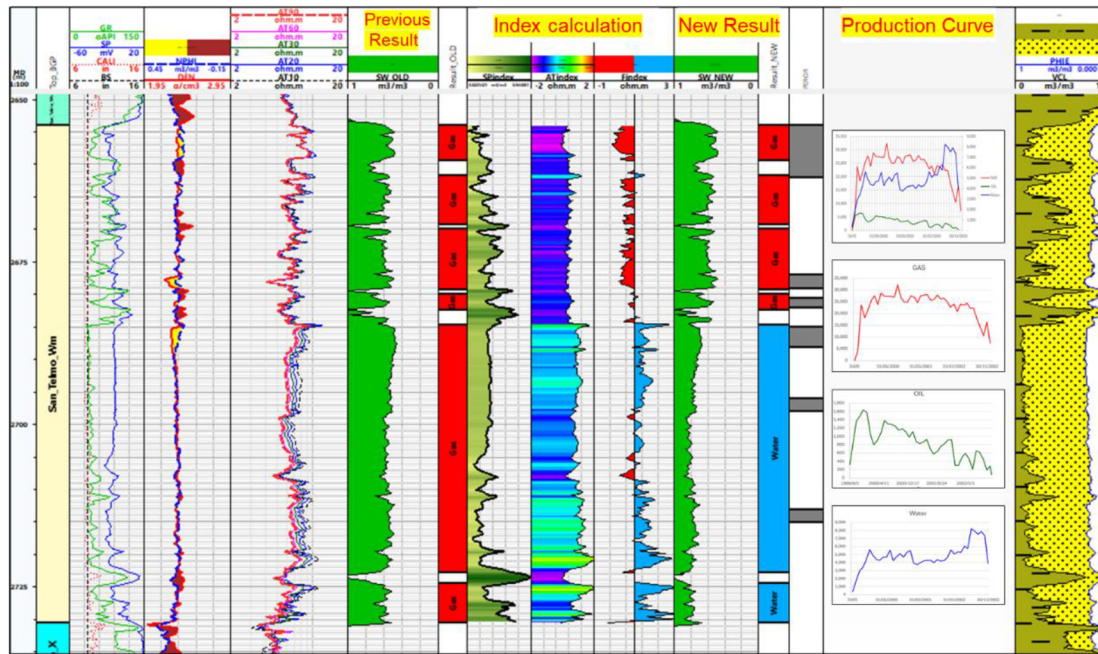


Figure 5—Validated with dynamic data, this approach enables fluid type, identified the upper low-resistivity hydrocarbon zones and the lower high-resistivity water layers, guiding exploration and development.

The initial log interpretation relied solely on resistivity (AT90), with both upper and lower sections showing similar resistivity values and comparable water saturation (Sw) calculations, leading to both zones being classified as gas-bearing. However, analysis of the resistivity amplitude difference between AT10 and AT90 revealed critical distinctions:

- Upper Section: Minimal resistivity amplitude difference ($\Delta AT = AT90 - AT10 \approx 0$), consistent with low-resistivity gas zones.
- Lower Section: Significant resistivity amplitude difference ($\Delta AT = AT10 - AT90$ and $AT10 > AT90$), combined with $R_{mf} > R_w$, indicative of a high-resistivity water layer (conductive invasion profile)

Through calculation by ATindex, SPindex and Findex, and analysis according to the cutoff value, the upper layer was a low-resistivity gas layer, the lower layer was a high-resistivity water layer, which updated the well log interpretation.

- Upper Section: Low-resistivity gas (ATindex <1, Findex <1)
- Lower Section: High-resistivity water (ATindex >1, Findex >1)

The production data from the perforated section provided critical validation of the updated log interpretation method. The observed high water cut of 54% strongly supported the reclassification of the lower zone as water-bearing, confirming the method's accuracy in fluid identification. Simultaneously, consistent gas production from the upper zone verified the presence of low-resistivity gas pay. These dynamic results demonstrate the superior reliability and practical value of the integrated SP-AT-Findex approach, particularly in resolving complex fluid distributions. The method demonstrated 92% accuracy versus production data (vs. 60% for initial interpretation), guiding exploration and development. The case proves the method's effectiveness in reducing development risks and optimizing perforation strategies, with the revised interpretation showing a perfect match with actual production performance. This successful field application highlights the technique's potential for widespread implementation in similar challenging reservoirs.

Conclusions

- This study proposes a novel fluid identification method. The technique integrates SP (Spontaneous Potential) and AT (Array Induction Resistivity) data, amplifying fluid differentiation through parameter computation to effectively distinguish high-resistivity water layers, low-resistivity gas zones and water-flooded intervals.
- The resistivity contrast between mud filtrate (R_{mf}) and formation water (R_w) forms the foundation for qualitative analysis, where fluid identification is achieved by amplifying differences through their distinct salinity characteristics.
- The key technical highlights of this method contain: Dual-data synergy- SP provides salinity sensitivity while AT delivers multi-depth resistivity profiles. Amplified fluid contrast-mathematical transformation enhances subtle differences imperceptible in single-log analysis. Three-dimensional differentiation: -solves classical interpretation challenges where conventional methods fail.
- Based on the latest fluid identification methodology, the updated well log interpretation has been validated through drilling and production data, providing reliable guidance for subsequent field development.
- This method is applicable to sand-shale sequences logged with the AT logging suite. Its underlying principle relies on the contrast between R_{mf} (mud filtrate resistivity) and R_w (formation water resistivity). It is particularly suitable for oil and gas fields drilled with fresh water-based mud and containing highly salinity formation water. The greater the resistivity difference between these two parameters, the more effective the fluid type identification.

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